BarCat Deals: Test and Validation Plan Draft

Course: CS 5001 - Senior Design I, University of Cincinnati
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Version: 0.1 | **Date:** November 4, 2026 | **Status:** AI-assisted draft; team review pending
QA/test lead: Kaustubh Mathur, following Week 8 task 02 and milestone F08; component owners review their cases.

1. What is being tested this cycle

This draft plans CS 5002 execution against Requirements v0.2 and Technical Specification v0.4, supported by User Stories and D0/D1/D2 v1.2. Current contracts supersede historical API-only collection, bookmarks, and active-only defaults. No implementation or passing results are claimed.

In scope	Requirement IDs	Component / primary owner
Today feed, facts, freshness, filters, accessible feedback	REQ-01, 03, 04, 08; NFR-01, 04	C1/G2: Brian Nguyen; C3: Ved Sanap
Approved-owner publication, validation, safe retry, visibility	REQ-05, 06; NFR-02	C2: Kaustubh Mathur; C3/C4: Ved/Meredith
Guest age, owner authority, service boundaries, health	REQ-02, 10; NFR-03; CON-03, 05	C3/G1: Ved Sanap; C4/G4: Meredith Bartel
Conditional approved-page scraping, extraction, recovery, spending	REQ-07; NFR-05; CON-01, 02, 04	C5/C6 boundary/G3: Kiki Vasilev
Database backup/restore and bounded logging	NFR-06, 07	C4/G4: Meredith Bartel

Major requirements: all nine Must requirements (REQ-01-05, NFR-01-04), plus REQ-06/07 and NFR-05 because duplication and unsafe automation threaten data integrity. Each has three cases, including a boundary or failure. All other in-scope IDs have at least one case.

Out of scope: REQ-09 active mode and NFR-08 100-times load are deferred Could features; TC-009/018 remain reserved. Won't features: bookmarks, moderation UI, email/SMS/push, self-service venue claiming, mandatory student accounts, DOB/ID images, walking routes, and access-control bypasses. Test our Supabase/Azure/Anthropic/source integrations, not provider internals or global availability. There are no sensors. Limits: self-attestation cannot verify age; radius is straight-line; fixtures cannot prove real-world coverage. Live scraping/AI requires OQ-01/04 approval.

2. Verification and validation strategy

Verification: component owners use pytest, real PostgreSQL integration tests, browser automation and axe-core. Test age/authority and atomic writes first, then geography/time, faults, performance and presentation. Unit tests isolate dependencies; integration tests exercise real constraints, concurrency and restores. Passing stubs cannot establish live-service acceptance.

Validation: recruit two students (one choosing the under-21 flow), an approved-owner representative and a keyboard/screen-reader user. Students locate and explain price/time/source/freshness (TC-003A/B, 004C); the owner submits and observes a deal (TC-005A, 012A); the assistive-technology user runs TC-014C. Target unaided completion of every listed task. Record completion, misunderstandings and accept/revise decisions against US-01-04; ask the advisor to review results/policies. This proposed small-sample protocol adds no approved NFR. Recruitment and sign-off are pending.

Layer	Approximate checks after parameterization	Coverage / execution point
Unit	80	Validation, occurrences, distance, age, matching, redaction; each commit
Integration	30	I1-I5, real C4 transactions, sessions, leases, budgets; each pull request
System	12	Integrated browser flows, faults, accessibility, timing, restore; milestone candidate
Acceptance	4 walkthroughs	Student, nonalcohol, owner, assistive-technology needs; before advisor/demo sign-off

Counts are estimates; cases expand into checks and may span layers. Section 5 fixes AI variables, metrics and trials; confidence/schema validity alone cannot establish factual accuracy.

3. Measurable success criteria

Compare unrounded values; a boundary equal to a strict upper limit fails. Each metric is tied to the Week 9 baseline. Proposed operating settings require recorded approval before acceptance.

Metric	Target threshold	Measurement / requirement
Eligible feed completeness and order	100% expected IDs, no extras/duplicates; distance <=1,609.344 m; start then ID order	Independent occurrence/distance oracle across all pages; REQ-01; TC-001A-C
Age and protected access	0 unauthorized alcohol disclosures; 100% tested unauthorized/CSRF-invalid actions denied; 0 secret/email leaks	Response bodies, HTML, browser state, logs, transport inspection; REQ-02; NFR-03; TC-002A-C, 013A-C
Facts, sources, freshness	100% cards/details have required labels; exact provenance; stale only >24 h, null = Never verified; no stale-only removal	Fixture assertions, all-page summary oracle; REQ-03/04; TC-003A-C, 004A-C; price acceptance gated by OQ-03
Owner validation and atomicity	7/7 omission trials name the item and commit 0 deals; valid input commits exactly 1 before success	UI/API responses and C4 before/after counts; REQ-05; TC-005A-C
Retry safety	<=1 deal across retries/concurrent copies; unchanged key replays original response for 24 h; altered body/new-key duplicate =409	Response comparison and row counts; REQ-06; TC-006A-C
Automated result integrity	One post/platform/external ID; 0-10 deals/post; complete validated facts and confidence >=0.70 to publish; .40-<.70 held, <.40 rejected	Fixed model stubs and database/provenance inspection; REQ-07; TC-007A-C
AI factual accuracy (proposed)	>=95% correct published deals; 0 wrong published alcohol, time, or price facts; zero publication cannot pass	100 unseen labeled posts, exact factual comparison, report publication coverage and omissions; REQ-07, CON-01; OQ-04; TC-007C, 019
Feed performance	>=95/100 initial loads <3 s with 5 concurrent sessions, 50 venues / 300 published deals	Browser navigation-to-complete-first-page-card-render, 20 batches of 5 cold visits; NFR-01; TC-011A-C. Also report 100/100 proposal under OQ-07 separately
Publication visibility	Every eligible fixture <=60 s from submit to visible card on new and already-open visible feeds	One browser monotonic clock for submit + observation; include processing/network/render; NFR-02; TC-012A-C
Accessibility/usability	0 critical/serious axe-core findings per tested page/state; 100% defined keyboard tasks, 0 traps	WCAG 2.1 A/AA axe rules, saved report and manual checklist; NFR-04; TC-014A-C
Recovery and operations	<=3 processing attempts, <=1 AI call/attempt; 0 accepted stale completions; <=1 committed result/post; daily backups, >=7-day retention, 1 successful isolated restore	Fault traces, AI-call counters, row counts, backup policy and restored integrity; NFR-05/06; TC-015A-C, 016
Health, logs, controls	Live=200; ready=200 only with mandatory settings/DB, else 503; buffer <=1,000, retention <=14 days, exact drop count; 0 unfunded/unapproved calls	Configuration/fault injection, clock advance, network/DB access probes; REQ-10; NFR-07; CON-01-05; TC-010, 017, 019-023
Filters and pagination	Exact expected sets; limit 1-50, default 20; 0 eligibility bypass; invalid/altered/expired/mismatched cursor=422	Compare all pages at limits 1/20/50; signed cursor expires at five minutes; REQ-08; TC-008

4. Test cases

Use the five columns below. TC-001A means case A in existing Week 9 group TC-001; no group is reassigned. All 47 cases are **Not Run** because implementation/execution is future work. On execution use only Pass, Fail, Blocked, or Not Run, with evidence under Section 6. Bracketed U/I/S/A identify unit, integration, system, or acceptance use. Fixture names and shared reset/capture steps are defined in Section 5; numbered actions are sequential. Requirements and D2 edge-case IDs appear in descriptions.

4.1 Core browsing and owner publication

ID	Description	Steps	Expected Result	Status
TC-001A	REQ-01 today set/order [I/S]	1. Reset F at T0; confirm age=true. 2. GET today for 2026-11-04 at limit=1 until cursor=null. 3. Compare API IDs and UI next pages.	Exactly C,A,G,B,F in that order; later-today B/F and previous-day overnight C included; D/E absent; no repeats.	Not Run
TC-001B	REQ-01 radius/effective-day edges; EC-05, EC-06 [U/I]	1. Evaluate R at the four recorded distances. 2. Query C on Nov 4 with final valid start date Nov 3. 3. Move its final date to Nov 2; repeat.	First three distances included, 1,609.345 m excluded; C included before change and excluded after. End on final occurrence, not midnight of its start day.	Not Run
TC-001C	REQ-01 DST boundaries; EC-07 [U/I]	1. Reset DST fixture clock to each test date. 2. Construct Mar 14 02:30-04:00 and Nov 7 01:30-01:45 occurrences in 2027. 3. Compare stored/returned UTC intervals.	Spring interval 07:00-08:00Z; fall interval 05:30-06:45Z; each returned once in selected-day feed. No nonexistent local instant.	Not Run
TC-002A	REQ-02 absent/declined/dismissed age; EC-01 [I/S/A]	1. Reset F and use fresh browser for each of no session, declined, dismissed. 2. Browse feed/all pages and alcohol detail F. 3. Explicitly request include_alcohol=true.	Feed contains C,A,G,B, 0 alcohol-offer content/IDs; food retained; detail and explicit alcohol request=403; browsing needs no registration.	Not Run
TC-002B	REQ-02 server authority [I/S]	1. With age=false, forge age/role in query, JSON and browser storage. 2. Request alcohol feed/detail. 3. Confirm via valid CSRF-protected age endpoint and repeat.	Forgery discloses 0 alcohol and explicit requests=403; valid server confirmation permits F; no DOB/ID requested.	Not Run
TC-002C	REQ-02 expiry/revocation; EC-01/15 [I/S]	1. Display F with confirmed session. 2. Repeat independently with expired and revoked token. 3. Trigger visible-feed revalidation, then request detail directly.	Invalid supplied token=401; revalidation removes alcohol cards and offers session renewal; 0 alcohol in failed response or refreshed DOM.	Not Run
TC-003A	REQ-03 facts/source [S/A]	1. Reset F. 2. Open owner A and automated B cards/details. 3. Read venue, offer, price, occurrence and check time; follow B's local HTTPS source stub.	All five fact labels match F; A says Submitted by venue; B links exactly its source and matching post, with no invented fact.	Not Run
TC-003B	REQ-03 missing price; OQ-03 [S/A]	1. Remove A's optional price. 2. Open its card/detail. 3. Submit extraction stub missing price for B.	Owner shows Price not provided; no guessed value; automated missing-price candidate hidden. Technical check can pass; full price-policy acceptance remains blocked pending OQ-03.	Not Run
TC-003C	REQ-03 unsafe content [I/S]	1. Put literal script/HTML in A's description. 2. Send automated source URLs with http: and javascript: schemes. 3. Open rendered detail.	Description rendered as text, 0 script executions; unsafe sources rejected/held with 0 public unsafe links; valid HTTPS provenance preserved.	Not Run
TC-004A	REQ-04 24-hour boundary; EC-05 [U/S]	1. Set A/B/C checks to T0-24 h, T0-24 h-1 s, and null. 2. Browse at T0. 3. Read labels and returned IDs.	A fresh; B Possibly out of date; C Never verified; all otherwise eligible IDs remain present.	Not Run
TC-004B	REQ-04 all-page metadata [I/S]	1. Give A/G old checks and B the newest check T0. 2. Query F limit=1 through all pages. 3. Compare summary with independent full-set oracle.	Every page reports same full-set last update T0 and stale count, even when newest B is on a later page; no per-page summary substitution.	Not Run
TC-004C	REQ-04 empty/unverified; EC-08 [I/S/A]	1. Query empty fixture. 2. Set every eligible check to null and repeat. 3. Explain each above-feed notice to observer.	Empty success: data=[], cursor/update=null, stale_count=0. All-unverified: update=null, count equals matching deals; notice states never verified; no false fresh/successful-update claim.	Not Run
TC-005A	REQ-05 seven-item success [I/S/A]	1. Reset O; sign in approved owner OA. 2. Submit valid seven-item form with key K. 3. Read committed deal/result rows and UI.	201 only after commit; exactly one deal, one owner subtype and request result; approved mapped venue VA; success notice identifies committed deal.	Not Run
TC-005B	REQ-05 missing/invalid fields; EC-02/07 [U/I/S]	1. Reset before each of seven logical-item omissions. 2. Repeat with description length 1,001, weekday 0 and equal times. 3. Submit UI and direct API.	Every invalid trial=422 at API with named text field error and 0 new deals; UI preserves other inputs. False alcohol is a valid value, not an omission.	Not Run
TC-005C	REQ-05 authority/rollback; EC-04/08 [I/S]	1. As OA send forged venue VB, then mismatched profile. 2. As nonowner attempt submit. 3. Inject DB failure before commit of valid form.	Forged/mismatched input=422, role=403; DB failure=503; 0 committed deals/subtypes/request results in each trial; no UI success, entered form retained.	Not Run

4.2 Retry safety and Must quality requirements

ID	Description	Steps	Expected Result	Status
TC-006A	REQ-06 unchanged-key replay; EC-03/09 [I]	1. Submit O with K. 2. Drop response; retry identical body/K at +1 s and +23h59m59 s using valid owner session. 3. Compare responses/rows.	Original status/body/ID replayed; exactly 1 deal and original result; no second publication.	Not Run
TC-006B	REQ-06 conflicts [I]	1. Commit K. 2. Change description with same K. 3. Send original duplicate under new key.	Both=409; exactly 1 deal; original result unchanged; error distinguishes conflict without overwriting facts.	Not Run
TC-006C	REQ-06 concurrent duplicate; EC-10 [I]	1. Release 10 identical OA submissions simultaneously, same K. 2. Repeat with 10 distinct keys for same offer. 3. Count rows.	One deal per trial; same-K responses replay same result; new-key losers=409; 0 partial subtypes.	Not Run
TC-011A	NFR-01 representative latency [S]	1. Reset P; apply Section 5 network/reference staging. 2. Run 20 batches of 5 concurrent cold feed visits, closing contexts each batch. 3. Save 100 complete-render durations and errors.	At least 95/100 <3,000 ms; all successful pages show full first-page cards. Errors/incomplete renders count as slow failures; report max and all samples, not API-only time.	Not Run
TC-011B	NFR-01 denied dependencies; EC-08 [S]	1. Repeat latency trial setup. 2. Stop API, then separately DB, and attempt 5 visits each. 3. Restore and rerun TC-011A.	Each failed visit has explicit error/retry and no empty-success claim; no failed sample discarded. Acceptance needs a new complete passing 100-visit run after repair.	Not Run
TC-011C	NFR-01 exact threshold [U/S]	1. Feed timing evaluator 94, 95 and 100 samples <3,000 ms out of 100; all remaining =3,000 ms. 2. Apply evaluator to raw TC-011A run; report stricter OQ-07 count.	94 fails; 95/100 pass baseline; 3,000 ms is not under 3 s. 100/100 is reported separately; no silent replacement of either acceptance rule.	Not Run
TC-012A	NFR-02 new-feed visibility [S/A]	1. Reset O with currently eligible nonalcohol deal. 2. Start browser clock at Submit; post form. 3. From second page in same browser, poll new feed navigations every second until card fully renders.	Committed ID visible <=60,000 ms after submit in each of 10 trials; no precommit visibility or success. Polling cost is included.	Not Run
TC-012B	NFR-02 already-open worst phase [S]	1. Keep student feed visible. 2. Submit just after a scheduled refresh starts. 3. Repeat 10 times with configured refresh and shaped network; time to automatic card render.	Each of 10 <=60,000 ms without manual reload. Record actual interval; a 60-second refresh plus processing cannot be excused by configuration.	Not Run
TC-012C	NFR-02 failure/retry; EC-03/08 [I/S]	1. Drop owner response after commit. 2. Retry same key/body. 3. Observe open and new feeds as in A/B, starting time at original Submit.	Original committed ID replayed; exactly one deal; visibility <=60,000 ms from original submit in 10/10 trials; retry does not restart timer.	Not Run
TC-013A	NFR-03 unauthorized matrix; EC-04, EC-15 [I]	1. For each protected owner/worker route send absent, forged and wrong-role credentials. 2. Repeat session routes with expired/revoked sessions. 3. Query DB changes.	All tested unauthorized actions denied (401 invalid authentication; 403 insufficient role/authority); 0 unauthorized mutations/disclosures; cross-venue body/profile validation separately covered by TC-005C.	Not Run
TC-013B	NFR-03 CSRF/transport [I/S]	1. Send every browser mutation with missing/wrong CSRF and cross-site origin. 2. Inspect cookies and HTTPS boundaries. 3. Probe plain HTTP.	100% invalid-CSRF mutations=403 before write; Secure/HttpOnly/SameSite=Lax cookies; plain HTTP redirects to HTTPS or is refused before sensitive content; no credentials sent over HTTP.	Not Run
TC-013C	NFR-03 privacy/redaction [I/S]	1. Seed unique canary token/key/email. 2. Exercise success, validation, login and DB/AI errors. 3. Search public JSON/HTML/logs and inspect no-store headers.	0 canary secrets/emails in public deals/logs; 0 raw posts/images in logs; Cache-Control: no-store on deal/session responses; errors retain request ID without secrets.	Not Run
TC-014A	NFR-04 loaded pages [S]	1. Load F feed and A/B details in reference browser. 2. Run pinned axe-core WCAG 2.1 A/AA rules after render. 3. Save per-page reports.	Every page has 0 critical and 0 serious violations; unresolved incomplete checks receive manual disposition; scan failure is not a clean result.	Not Run
TC-014B	NFR-04 adverse states; EC-01/08 [S]	1. Render empty, restricted, stale and API-error feed/detail states. 2. Run same axe configuration on each. 3. Inspect notices/focus.	Each state has 0 critical/serious findings; notices readable without color alone; no hidden alcohol exposed in restricted state.	Not Run
TC-014C	NFR-04 keyboard/screen reader [S/A]	1. Unplug mouse; use Tab/Shift-Tab/Enter/Space to dismiss age, select food, follow Next, open detail, return. 2. Repeat with screen reader reading facts/source/stale notice. 3. Record each task.	5/5 navigation tasks complete, 0 traps; visible focus and meaningful names; venue/offer/price/time/check/source announced correctly. No observer intervention for task completion.	Not Run

4.3 Supporting requirements, recovery, and constraints

ID	Description	Steps	Expected Result	Status
TC-007A	REQ-07 normalization/limits [U/I]	1. Register same normalized post twice and with another platform. 2. Complete posts with 0, 10 and 11 candidates; test text 5,001 chars/11 images.	Same platform/ID registered once; other platform distinct; 0 means no_deal, 10 valid allowed; oversized inputs=422/held, 0 partial invalid-batch publications.	Not Run
TC-007B	REQ-07 confidence/validation; EC-12 [U/I]	1. With approved test settings, complete valid stubs at .3999/.40/.6999/.70. 2. Repeat .99 with invented/invalid date, missing price and ambiguous alcohol.	Thresholds reject/hold/hold/publish respectively; malformed/missing facts never published; uncertain alcohol=true; raw post and provenance retained. A high score cannot bypass validation.	Not Run
TC-007C	REQ-07 factual evaluation [I/A]	1. Freeze H labels/model/prompt/settings. 2. Run each of 100 posts once; two reviewers compare published candidates to evidence. 3. Calculate Section 3 accuracy and coverage; keep publisher off until approved.	>=95% fully correct published deals and 0 wrong alcohol/time/price; 0 published fails; report held/rejected/missed counts. OQ-04 approval needed; no score-based accuracy claim.	Not Run
TC-008	REQ-08 filters/cursors; EC-07 [I/S]	1. Query F by venue and each category at limits 1/20/50. 2. Try 0/51 and altered, five-minute-expired, changed-filter cursors. 3. Repeat age=false.	Exact oracle sets; food filter prominent; default20; invalids=422; no duplicates, widened radius, or unauthorized alcohol.	Not Run
TC-010	REQ-10 health; EC-08 [I]	1. Probe live/ready normally. 2. Remove mandatory campus setting, then restore and deny DB separately. 3. Inspect bodies.	Normal both200; each fault ready503/live200 while process running; no secrets; recovery restores ready200.	Not Run
TC-015A	NFR-05 crash/lease loss; EC-11 [I]	1. Claim post, kill worker before completion. 2. Advance past 120 s lease; reclaim. 3. Submit old token, then current result.	Old completion=409; only current lease can commit; exactly1 stored result; raw post survives crash.	Not Run
TC-015B	NFR-05 retry limits; EC-13 [I]	1. Stub AI timeout on every call. 2. Run scheduler through all retries with controlled clock. 3. Count calls/attempts; inspect failed post.	<=3 attempts, <=1 AI call each; 30/120 s delays plus 0-10% jitter; after third failure raw retained, no fourth call/publication.	Not Run
TC-015C	NFR-05 completion replay; EC-09/10 [I]	1. Commit valid completion then lose response. 2. Retry same completion; race two workers on matching offers. 3. Inject deadlock twice.	Replay identical result; <=1 published offer with preserved evidence; one deadlock retry then503, no partial commit; late/replaced tokens409.	Not Run
TC-016	NFR-06 isolated restore [I/S]	1. Inspect >=8 days of scheduled backup evidence. 2. Restore latest backup into isolated DB. 3. Compare snapshot counts/checksums, foreign keys and provenance; run migration-failure rollback trial.	Backup gap <=24 h, retention >=7 days; one restore succeeds with 0 FK/provenance/checksum differences; failed migration leaves prior version usable; no production overwrite.	Not Run
TC-017	NFR-07 collector outage [I]	1. Deny collector; emit numbered events 1-1,205; browse. 2. Restore collector. 3. Advance retention clock beyond14 days.	Buffer never>1,000; retained 206-1,205; drop count205 reported; feed remains usable; no events older than14 days retained.	Not Run
TC-019	CON-01 funding gate; EC-13 [I/A]	1. Disable funding approval/model, then test cap with concurrent reservations crossing remaining balance. 2. Deny AI budget. 3. Submit owner deal.	0 unapproved/unfunded calls; reserved total<=approved cap, duplicate attempt reserves once; denial consumes0 attempts; owner publication works. OQ-04 approval required to enable AI.	Not Run
TC-020	CON-02 inclusive fallback; EC-14 [S/A]	1. Use owner venue with no social account. 2. Disable all external sources/AI. 3. Complete TC-005A/012A.	Owner can submit without social login/post; eligible deal visible<=60 s; source says Submitted by venue.	Not Run
TC-021	CON-03 food/minimal data [I/S]	1. Browse fresh/declined session; select food. 2. Inspect UI/API/session schema and recorded data.	Food filter prominent; exact eligible food set,0 alcohol; no DOB/ID field, upload, or stored value; only age answer stored.	Not Run
TC-022	CON-04 permitted scraping; EC-14 [I/A]	1. Attempt unapproved URL, then approved local stub returning429 with Retry-After60,403, and500. 2. Advance clock and inspect calls. 3. Review one permitted live fetch per enabled adapter.	0 unapproved calls; no429 retry before60 s (15min fallback if absent);403 stops adapter;500 retries next permitted poll; no bypass; missing live-access evidence blocks enablement.	Not Run
TC-023	CON-05 DB isolation [I/S]	1. Inspect deployment identities/env references. 2. From C1/C2/C5 attempt direct C4 connection; call C3 instead. 3. Review migration route.	Only C3 has application DB credentials/network access; all direct client/worker probes denied; I1-I3 work; migrations use controlled C3 deployment identity.	Not Run

5. Test environment

Section 4's sample datasets are called **fixtures**. **Reset** means reload a fresh copy. **T0** is November 4, 2026, noon Eastern (17:00 UTC). Test software/data still need to be built.

Where tests run

Environment	Setup and purpose
Development	Local Flask/Jinja/FastAPI/Pydantic, PostgreSQL matching staging, pytest, browser tools and axe-core. Separate test database; simulated source/AI/login services. Live scraping/paid calls default off.
Staging	Azure web/API/worker containers, managed PostgreSQL, HTTPS, production WSGI server, Supabase test accounts. Proposed: one API instance (2 vCPU/4 GiB), one web instance, one worker. Record database tier; fix resources during timing. Funding/approval required; no production data.
Pilot	Invited students/owners use the approved staging build after Must tests pass. Record devices/browsers and setup differences. AI stays off unless approved.

Record the setup: at S01, Meredith/Ved save hardware, exact software/container/migration/timezone versions, code/data versions, dataset checksum and nonsecret settings. Missing details block affected tests. Draft reference: Windows x64 25H2 build 26200.9711, Edge 156.0.4314.58; proposed Python 3.12/PostgreSQL 16. Confirm versions at S01. Use >=4 logical cores/8 GB RAM, keyboard, NVDA screen reader and 390x844 mobile emulation; emulation cannot prove physical-device compatibility.

Before each test: reload/check the dataset; set server/worker clocks; reset sessions, budgets, counters and simulated services. Save requests/responses, screens and database changes. Clock/failure controls are test-only. Use `/api/v1` (worker: `/api/v1/internal` with service key), agreed field names and valid CSRF tokens except negative tests. At S01, document commands to run cases by ID.

F - sample feed: “Reset F” loads the whole dataset; “deal F” is its alcohol offer. Use a fictional campus at (0,0), approved venues, invented details and matching source records. Default distance: 1,000 m; last checked: T0.

Deal ID	Source/type	Eastern schedule	Price / special condition
A	Owner / food	Wed 12:00-13:00	$5
G	Owner / food	Wed 12:00-13:00	$6; exactly 1,609.344 m away
B	Automated / food	Wed 18:00-20:00	$7; original post on simulated HTTPS source
F	Automated / alcohol drink	Wed 20:00-22:00	$3
C	Owner / food	Tue 23:00-Wed 02:00	$4; final allowed start November 3
D	Otherwise eligible deal	Overlaps selected day	1,609.345 m away; excluded
E	Otherwise eligible deal	Overlaps selected day	Pending review; excluded

For TC-001A, confirm age and request `include_alcohol=true`: **C,A,G,B,F**. Normal nonalcohol feed: **C,A,G,B**. IDs put A before G. Last update = newest known check across all matching deals; stale count uses that full set.

Other labels used in Section 4

Label	Meaning and data to use
R - radius	Distances: 1,000 / 1,609.343 / 1,609.344 / 1,609.345 m. Include first three; exclude last. Use exact numbers in unit tests; independently check integration coordinates within 0.1 mm, plus +/-1 cm trials. Real coordinates need OQ-02 approval.
DST - daylight saving time	New York clock changes: March 14 / November 7, 2027. Set clock to the selected day to stay within the 14-day query limit.
O - owner submissions	Owner OA owns only venue VA; owner OB owns VB. Include nonowner/expired/revoked sessions; K is a UUID submission key. Form: matching business name/address/email, description, category, weekdays/times, alcohol=false; Wed 12:00-13:00. Timing requires currently eligible offers.
P - performance	Repeatable 50 venues/300 published deals: both sources, food/alcohol, stale/overnight. Record historical volume; check an independently sorted CSV. Run 100 visits, 5 at once: 10 Mbps down/1 Mbps up/100 ms round-trip delay, no packet loss. Disable browser cache, warm services once, fix data/resources, save settings/timings. OQ-07 approval pending.
H - AI accuracy	Kiki prepares 100 unseen approved/anonymized or synthetic posts: 20 no-deal, 20 multi-offer, 20 unclear/missing facts, 40 single clear offers; include food/alcohol/flyers/overnight/reposts. Retain text/images/timestamps, source hashes, expected 0-10 offers. Kaustubh/Kiki independently agree answers; hide them from AI. Fix model/prompt/source format and supported temperature/seed; one call/post. Record cost/variation, correct published deals / all published deals, wrong alcohol/time/price, held/rejected/missed offers and coverage. Require >=95% correct published deals, zero wrong alcohol/time/price; publishing nothing fails. Stakeholders review omissions; no minimum offer-discovery rate is approved. Changed model/prompt needs new unseen data/approval. OQ-01/04 gate live access/funding.

6. Pass and fail criteria

Per case: Pass requires every trial/outcome to meet Section 3 on the frozen environment. Wrong/missing values, disclosure, duplicate/partial commits or breached thresholds are Fail. Blocked means missing prerequisites/decisions; Not Run means no attempt. Neither passes. Retain failures; rerun repaired cases and affected regression. Mocks cannot establish live acceptance.

Whole plan/release: 100% Must cases/all five constraints pass; 0 open critical/high defects. Critical: age/credential disclosure or data loss. High: broken browsing/submission, integrity or accessible-task failure. Selected Should cases pass or have a team/advisor exception naming impact, owner and disabled feature. Disabled automation is not tested coverage. Could items stay excluded; lesser defects need accepted dispositions.

Acceptance blockers: OQ-03: optional price versus AC-01.1 completeness. OQ-07:95/100 versus every-load<3 s, and specified 60-second versus proposed 30-second refresh within end-to-end 60 s. Record both latency counts and actual refresh/visibility results. OQ-01/04 gate live automation; OQ-02 real geography; OQ-06 constraint/plan reconciliation. October 30 review dates have passed without closure evidence. Technical tests may run, but affected acceptance/sign-off requires documented resolution; do not infer approval or weaken targets.

Evidence: record case/requirement IDs, date/operator, commit/manifest/seed hashes, expected/actual values, status, links, defect/severity, reviewer and retest. Planned path: `test-evidence/<milestone>/<run-id>/` for JUnit, screenshots, timing CSV/HAR, axe/manual/DB/AI reports and decisions. Remove secrets/contact emails/unapproved material. No results exist yet.

7. Test schedule

Dates/IDs follow Week 8 Planning; spring gates are proposed. Owners review cases; Kaustubh coordinates evidence, Ved release review. Activities remain planned.

Milestone / date	Test activity / Week 8 task linkage	Owner	Required evidence
F08 Nov 4, 2026	Draft scope, metrics and cases; task 02. Swap two cases with another team before freezing.	Kaustubh; all owners	Plan/PDF, independent execution feedback or pending action
F10/F11 Nov 18; F12 Dec 2	Review plan/DoD with advisor; five-test triage before presentation; carry plan into design-report appendix.	All; Kaustubh coordinates	Advisor decisions, open issues, linked report appendix; no invented execution
S00 Jan 11; S01 Jan 25, 2027	During tasks 06-10: unit validators/time/age first; real-DB contracts, fixture manifest, UI/form stubs; continuous commit/PR checks.	Meredith/Ved; each owner	Locked manifest, seed/oracle hashes, unit/JUnit and I1-I4 reports
S02 Feb 9	Tasks 11/14/16: auth, lease crash/retry, budget, staging isolation/health.	Ved/Kiki/Meredith	TC-002/006/013/015/019/023 fault traces and row counts
S03 Feb 24	Tasks 12/13/18: manual vertical slice and H evaluation; automation stays gated.	Brian/Kaustubh/Kiki	TC-003/005/007/012/020 results, source/quality evidence
S04 Mar 10; S05 Mar 24	Tasks 15/17-20: system boundaries,100-load timing, axe/keyboard; defect corrections and regression before risk review.	Ved/Brian; all owners	TC-001/004/008/011/014 results; defect/retest log
S06 Apr 5; S07 Apr 12	Tasks 21-23: restore, regression, student/owner/assistive walkthroughs before acceptance review.	Meredith/Kaustubh; Ved	TC-016 restore report; stakeholder task records, signed acceptance or blockers
S08 Apr 19; S09 Apr 28; S10 Apr 30	Tasks 24/25: automation decision, frozen release regression, second-operator handover and final demo.	Kiki; Meredith; all	Funded/access/AI approvals or explicit exception, final matrix, runbook and sign-off

Proposed five-test triage card: likelihood/impact1-3; rank their product, then core flow. Run before demos; full acceptance is still required.

Run order / case	Requirement	Check	Likelihood x impact = score
1 / TC-002B	REQ-02	Forged age cannot expose alcohol	3 x 3 =9
2 / TC-005C	REQ-05	Cross-venue/DB failure commits nothing	3 x 3 =9
3 / TC-001A	REQ-01	Complete today's set and order	3 x 3 =9
4 / TC-012B	NFR-02	Open feed sees submission<=60 s	3 x 3 =9
5 / TC-014C	NFR-04	Keyboard/reader completes core tasks	2 x 3 =6

TC-002B is done when forgery exposes zero alcohol, valid confirmation permits the offer, and a reviewer reproduces the evidence.

8. Definition of done

Feature checklist

[ ] Agreed requirement/acceptance IDs, design contract, owner and cases; no unapproved scope.

[ ] Code/migrations peer-reviewed by another teammate and merged through normal review.

[ ] Unit/integration checks pass, including boundaries, faults and relevant real-DB checks.

[ ] System results meet Section 3; UI success reflects committed data and correct authorization.

[ ] Accessibility/security pass; no open critical/high defect.

[ ] Reproducible hashes, actual outcomes and sanitized failure/retest evidence saved.

[ ] Traceability, API docs and user/runbook instructions current; changed IDs logged/mapped.

[ ] Applicable stakeholder validation recorded; unresolved policies remain explicit.

Project checklist

[ ] All nine Must requirements/five constraints satisfy Section 6; selected Should cases pass or have disabled-feature exceptions.

[ ] Requirements/specification reconcile OQ-02/03/07 acceptance; OQ-06 conflicts resolved.

[ ] Live automation has OQ-01 access and OQ-04 funding/model/quality approval, or an explicit manual-only exception.

[ ] Frozen-build regression, restore, logs, HTTPS, least privilege and second-operator handover evidenced.

[ ] Student/owner/assistive-technology validation recorded; feedback fixed/retested or accepted.

[ ] No open critical/high defects; other defects/limits/deferred scope have owners and accepted dispositions.

[ ] QA lead/component owners confirm evidence; advisor records dated accept/revise decision. Submission is not sign-off.

[ ] Final-report appendix/README link source/PDF/results; all five names included; team owns reviewed AI-assisted metrics/cases/DoD.

Current state: all cases Not Run; harness, environment freeze, recruitment, team review and advisor acceptance pending.

Appendix A. Supporting sources

Source	Use in this plan
Requirements v0.2, Sections 5-9	Requirement IDs, priorities, targets, assumptions and open decisions
Technical Specification v0.4, Sections 2-6	Interfaces, data formats, limits, failure behavior and test mappings
D0, D1, D2 v1.2	Component ownership, scheduling, transactions, recovery and EC-01-15
User Stories; Constraints; Project Description	Stakeholder needs, economic/social/ethical limits and current scope
Planning Package; Task List	Milestone dates, implementation tasks and owners
Peer Review Notes, PR-01-05	Coverage, source access, day semantics, references and extraction-quality risks; closure not assumed

Appendix B. Deferred Could procedures

These procedures preserve Week 9 IDs and run only after a recorded scope decision.

ID	Description	Steps	Expected Result	Status
TC-009	REQ-09 deferred active mode [U/I]	1. Only if scope approved, query 12:00-13:00 offer at 11:59:59/12:00/12:59:59/13:00. 2. Compare food-first/minutes/ID ordering.	Absent/present/present/absent; remaining minutes rounded up; exact tie order. Excluded from this cycle's completion claims.	Not Run
TC-018	NFR-08 deferred stretch [S]	1. Only if approved, seed 5,000 venues/30,000 published/about 5 million historical rows. 2. Repeat TC-011A.	>=95/100 loads<3 s at 5 concurrent with same network; failure does not fail MVP; report actual volume.	Not Run

Coverage: 21 IDs in scope; REQ-09/NFR-08 deferred. All 23 original groups retained:36 major cases (12x3),9 other in-scope cases,2 deferred procedures =47. D2 EC-01-15 are covered.

Document links: Markdown source and PDF. Team review and ownership of the metrics, cases and definition of done remain pending.